

The Competition-Independence Game in Trees

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ABSTRACT. The competition-independence number of a graph is defined by the following game. Two players take turns in constructing a maximal independent set M , where one player tries to maximize $|M|$ and one player tries to minimize $|M|$. Phillips and Slater determined the parameter for paths. In this note, we first revisit paths and determine the parameter that results when maximal independent set is generalized to maximal k -packing. Then we provide bounds for the original parameter on trees of maximum degree at most 3.

Dedicated to Pete Slater, a great friend who helped us, already as students, with his wit, wisdom and generosity.

1 Introduction

Phillips and Slater [4, 5] introduced what they called the ***competition-independence game***. The game is played by two players, Diminisher and Sweller, on some graph G . They take turns in constructing a maximal independent set M of G . That is, each turn a player chooses a vertex that is not adjacent to any of the vertices already chosen until there is no such vertex. The goal of Diminisher is to make the final set M as small as possible and for Sweller to make the final set M as large as possible.

This game is related to some other games on graphs. For example, our interest was rekindled by the domination game, introduced by Brešar et al. [2], and variations thereof for other domination parameters (see for example [3]). The game is also related to (generalized) Kayles (see for example [6]): there the players similarly construct a maximal independent set, but the winner is determined by the parity of the resulting set. There are also other games on graphs with a maximizer and a minimizer; see for example [1, 7].

For a graph G , let $I_d(G)$ denote the optimal size of the final independent set in the competition-independence game if Diminisher moves first. Similarly, let $I_s(G)$ denote the optimal size of the final independent set if Sweller

moves first. We call these the **competition-independence** numbers. Note that these numbers can vary greatly on who goes first. For example, if the graph G is a star, then if Diminisher goes first he will choose the center vertex. In contrast, if Sweller goes first she chooses an end-vertex, and so the set M will end up with all but one vertex. That is, $I_d(K_{1,m}) = 1$ while $I_s(K_{1,m}) = m$. In contrast, well-covered graphs—where every maximal independent set has the same size—provide games where the players’ moves don’t matter.

Philips and Slater [5] showed that

Theorem 1 *The competition-independence numbers of the path on n vertices are given by:*

$$I_d(P_n) = \lfloor (3n + 4)/7 \rfloor \quad \text{and} \quad I_s(P_n) = \lfloor (3n + 6)/7 \rfloor.$$

The above theorem immediately provides the value for an n -cycle, since the first move in C_n produces P_{n-3} .

In this paper we build on Theorem 1 in two directions. In Section 2 we generalize it to what we call the competition-packing game. Then in Sections 3 and 4 we investigate the maximum and minimum values of the competition-independence game for trees of maximum degree 3. We conclude with some general thoughts about trees.

2 Paths and Packings

Consider the **competition- k -packing game**. In such a game, the two players create a maximal set with the property that no two vertices are within distance k of each other. Equivalently, this is the competition-independence game played on the k^{th} power of the graph. For a graph G , let $I_d^k(G)$ denote the optimal size of the resultant k -packing if Diminisher moves first, and let $I_s^k(G)$ denote the optimal size if Sweller moves first.

The upper and lower k -packing numbers of the path on n vertices are asymptotically $n/(k + 1)$ and $n/(2k + 1)$ respectively. We show here that the competition- k -packing number of the path on n vertices is $3n/(4k + 3)$ for n a multiple of $4k + 3$. Specifically: for $1 \leq r \leq 4k + 3$, define

$$\delta(r) = \begin{cases} 1 & \text{if } 1 \leq r \leq 2k + 1 \\ 2 & \text{if } 2k + 2 \leq r \leq 3k + 2 \\ 3 & \text{if } 3k + 3 \leq r \leq 4k + 3 \end{cases} \quad \& \quad \sigma(r) = \begin{cases} 1 & \text{if } 1 \leq r \leq k + 1 \\ 2 & \text{if } k + 2 \leq r \leq 2k + 2 \\ 3 & \text{if } 2k + 3 \leq r \leq 4k + 3 \end{cases}$$

We say a vertex u **covers** a vertex v if they are within distance k of each other.

Theorem 2 For $k \geq 1$, the competition- k -packing numbers of the path on n vertices, where $n = (4k + 3)q + r$ with $1 \leq r \leq 4k + 3$, are given by:

$$I_d^k(P_n) = q + \delta(r) \quad \text{and} \quad I_s^k(P_n) = q + \sigma(r).$$

PROOF. (1) We establish the upper bound for both versions by providing a strategy for Diminisher.

Say $n = Cq + r$ with $1 \leq r \leq C = 4k + 3$. Then Diminisher mentally partitions P_n into q tracts of C vertices and one tract of r vertices. For each C -tract, mentally color red both vertices that are at distance k from an end-vertex of the tract. Define a C -tract as *live* if only one vertex has been chosen from it and that was by Sweller. Note that if the C -tract is live, then at least one of the two red vertices must be choosable. (Neither can be covered by a vertex outside the tract, and they are too far apart to be both covered by Sweller's initial vertex in the tract.)

For the r -tract, if $r \geq k + 1$ and $q > 0$, then color red the vertex that is distance k from the end-vertex of the tract that has a neighbor outside the tract; if $r \geq k + 1$ and $q = 0$, then color red one vertex that is distance k from an end-vertex of the tract. If $r > 3k + 3$, then also color red the vertex that is distance- k from the other end-vertex of the tract. Define an r -tract as *live* if only one vertex has been chosen from it and that was by Sweller, and there is a red vertex in it that is choosable.

The strategy for Diminisher is as follows:

- If there is a live tract, then choose a red vertex there;
- Else choose a red vertex in the r -tract if possible;
- Else choose any red vertex if possible;
- Else choose an arbitrary vertex.

Note that this strategy ensures that there is at most one live tract at any stage.

We claim that: no more than three vertices are chosen from each C -tract. For, the strategy for Diminisher ensures that in every C -tract at least one red vertex is chosen. This covers $2k + 1$ vertices, and at most two of the remaining $2k + 2$ vertices can be chosen.

Further, consider the r -tract. If Diminisher moves first, he chooses a red vertex if there is one. The remaining choosable vertices are $2k + 2$ at most, so $\delta(r)$ vertices can be chosen from the tract. Consider Sweller going first. If $r \leq 3k + 3$, then at most $\sigma(r) = \lceil r/(k + 1) \rceil$ vertices can be chosen no matter what the players do. If $r > 3k + 3$, then Diminisher will choose a red vertex if possible in response to Sweller's first move. One can readily

verify that at most three vertices can be chosen from the r -tract, both if Diminisher cannot choose a red and if Diminisher can choose a red.

This proves that Diminisher can ensure that the resultant k -packing has size at most the stated quantity.

(2) We establish the lower bound for both versions by providing a strategy for Sweller. Let $f_s(n)$ denote the minimum competition- k -packing number over all linear forests (that is, union of paths) on n vertices with Sweller starting, where Diminisher is also allowed to pass (not make a move). Define $f_d(n)$ similarly but with Diminisher starting.

The strategy for Sweller is as follows:

- If there is a component with order at least $k+2$, then choose a vertex that is distance- $(k+1)$ from a leaf. Otherwise choose any vertex.

If every component has order at most $k+1$, then the game number is the number of components, which is at least $\lceil n/(k+1) \rceil$. This quantity is at least $q + \delta(r)$. If there is a component with order at least $k+2$, then Sweller's move creates an isolated vertex and covers at most $2k+1$ vertices. So we have $f_s(n) \geq 2 + f_d(n-2k-2)$. On the other hand, we have $f_d(n) \geq 1 + f_s(n-2k-1)$, since any move covers at most $2k+1$ vertices. Combining these inequalities yields that $f_s(n) \geq 3 + f_s(n-4k-3)$ for $n > 4k+3$ and similarly with $f_d(n)$. So it suffices to check the lower bound for $n \leq 4k+3$, which is readily verified. QED

3 Maximum Value in Trees of Maximum Degree 3

We return to the original competition-independence number. We consider here trees of maximum degree at most 3.

3.1 Small Orders and a Construction

It is easy to find trees with maximum degree 3 where the value of the game is half the order. For example, consider the corona of a path (that is, the caterpillar created by taking a path and adding a new end-vertex incident with each vertex of the path). Note that this graph is well-covered; so it follows that the game will produce a set of half the vertices, no matter how the players move.

For the case that Sweller goes first, it is easy to find trees of maximum degree 3 where the game value is bigger than half the order, such as $K_{1,3}$.

For another example, let U be the tree of order 10 and diameter 4 where the center and its three neighbors have degree 3. Then $I_d(U) = 4$ (first move is to choose a neighbor of the center), while $I_s(U) = 7$ (first move is to choose the center).

For the case that Diminisher moves first, computer search shows that $\lfloor n/2 \rfloor$ is the maximum value of the competition-independence number of all trees of maximum degree 3 up through order at least $n = 15$. However, this does not hold true for all n , as we now show.

We use the tree U to build a tree W where, even with Diminisher going first, the value is more than half the order. Take four copies of U and root each copy at an end-vertex. Say the copies are U_1, U_2, U_3, U_4 with r_i the root in U_i . Then identify r_1 with r_2 , identify r_3 with r_4 , and add an edge between them. The resultant tree W has order 38. See Figure 1, where the two roots are in white.

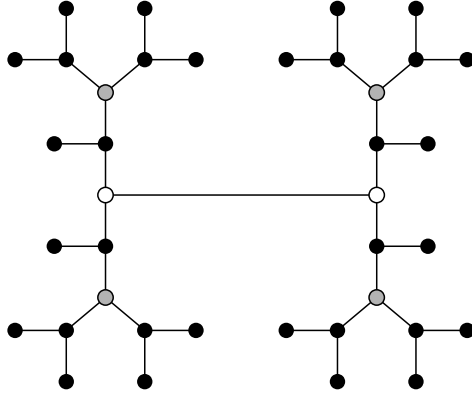


Figure 1. The tree W

We will need the following simple result:

Observation 1 *If graph H is obtained from graph G by adding two isolated vertices, then $I_d(H) = I_d(G) + 2$ and $I_s(H) = I_s(G) + 2$.*

PROOF. Both players can ensure that the stated value is a bound by following their strategy on G until the other player chooses the first isolate, at which point they choose the other isolate, and the other player is forced to continue on G . QED

We now establish the following:

Theorem 3 *With Diminisher going first, the competition-independence number on the above tree W has value $I_d(W) \geq 20$.*

PROOF. Consider the four copies of U . Any maximal independent set M in W contains at least three non-root vertices from each copy of U . If M contains the center of a copy of U (shown in grey in the figure), then at least six non-root vertices from that copy are in M .

Thus, if M contains three of the original centers, then M will have size at least $3 \times 6 + 3 = 21$. So we may assume that Sweller is not allowed to choose three centers. Sweller can, however, choose at least two centers on her first two moves. And so, Diminisher must choose neighbors-of-centers on at least two of his first three moves.

Thus, without loss of generality, we can assume Diminisher chooses neighbor-of-center on his first two moves, with Sweller choosing the centers in the other two copies of U .

So consider the situation with Diminisher to choose his third vertex. By Observation 1, we can set aside ten isolated vertices created by Sweller's first two moves, as well as the 16 vertices already dominated. Thus we are left with a forest on 12 vertices that includes at least two copies of P_3 . Let F denote the subgraph of the other six vertices. Then F is either $2P_3$, P_6 , or the star $K_{1,3}$ with two of the edges subdivided once. It remains to verify that $I_d(F \cup 2P_3) \geq 6$.

Sweller will choose in the $2P_3$ only when F is done or when Diminisher chooses there. When he chooses there, she will choose a leaf. Note that that allows Diminisher to choose there rather than in F . So it is sufficient to show that for all three possible F , even if Diminisher is allowed to pass, at least 3 vertices of F are chosen. This is easily checked. QED

The above idea can be generalized. Let tree W_m be obtained from $m + 2$ copies of U , say $U^0, \dots, U^{m+1}$, rooted at end-vertices, where the roots of U^0 and U^1 have been identified, the roots of U^m and U^{m+1} have been identified, and edges have been added so that the roots form a path. That is, W_2 is W in Figure 1. Then it can be shown that the value of the competition-independence game, with either player starting, is at least $(\frac{1}{2} + \varepsilon)n$ for some small ε , where $n = 10m + 18$. We omit the details.

3.2 Upper Bound

We are still considering trees of maximum degree at most 3. These trees have independence number at most $3n/4$. (If M is any independent set, then by counting the edges incident with M it follows that $|V(G) - M| \geq |M|/3$.) The upper bound for the independence number provides an upper bound on the competition-independence number. But we now improve on it:

Theorem 4 For a tree T of order $n \geq 2$ and maximum degree at most 3, with Diminisher going first, the value of the independence game $I_d(T) \leq 4n/7$.

PROOF. Let $\alpha = 4/7$. We prove that the maximal independent set M that is constructed has $|M| \leq \alpha n$. At each stage, let I be the vertices that have been chosen and J_I the isolates in $T - N[I]$. Define the *energy* as

$$\psi(I) = |I| + |J_I| + \alpha|V(T) - N[I] - J_I|.$$

Note that $\psi(\emptyset) = \alpha n$. Note further that if a vertex of J_I is chosen then the energy does not change. Otherwise, the change in energy is given by $(1+e)(1-\alpha) - d\alpha$ where d vertices are added to $N(I)$ and e vertices are added to J_I .

Let F denote the forest after an even number of moves. Then the strategy for Diminisher to ensure that $|M| \leq \alpha n$ is:

- choose in F the vertex that minimizes the resultant energy.

Say the move changes the energy by D . We will show that: *after the response from Sweller, the energy is at most what it was in F .*

Consider all possible forests that could result after *any* move by Diminisher. Out of all such forests and all possible responses by Sweller, consider the response that maximizes the change in the energy. Say this response is choosing vertex v in F' forming F'' ; let S be the change in energy. It suffices to show that $D + S \leq 0$.

Suppose first that $S \leq 0$. If vertices x and y in F are distance 3 or more apart, then choosing x cannot lower the change of energy that results from choosing y , and vice versa; it follows that $D \leq 0$. If some component of F is a star, then Diminisher can choose the center, and again $D \leq 0$.

So assume that $S > 0$. Let w be the neighbor of v adjacent to the most end-vertices, say C . Since $S > 0$ it follows that $C > 0$. (But also $C \leq 2$ by the maximum degree constraint.) Let D' be the change in energy when Diminisher chooses w in F . It suffices to show that $D' + S \leq 0$.

Now, consider the subgraph near v in F' . Say vertex v has A non-leaf neighbors and B leaf neighbors. If we maximize S without changing D' , then it follows that we may assume that each non-leaf neighbor of v has exactly C leaf neighbors. See Figure 2 overleaf.

It follows that, in F' , the choosing of v creates AC isolates while dominating the $A + B$ neighbors of v . Thus

$$S = (1 - \alpha)(1 + AC) - \alpha(A + B).$$

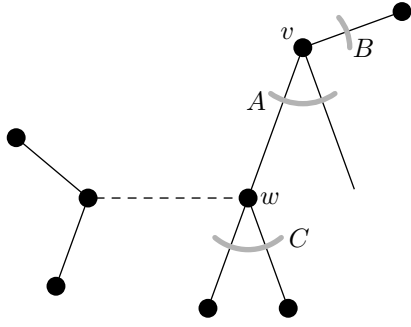


Figure 2. Possible choices for Sweller and Diminisher

Now, go back to the graph F . There could be at most one extra edge in our picture in F than in F' . To maximize D' we may assume that vertex w has degree 3, with any other non-leaf neighbor (if any) having 2 leaf neighbors. (This uses the fact that $2\alpha - 1 > 0$.) The choosing of w creates $B + 2(2 - C)$ isolates while dominating the 3 neighbors of w . Thus

$$D' = (1 - \alpha)(1 + B + 2(2 - C)) - 3\alpha.$$

It remains to verify that $S + D' \leq 0$ in all cases. The partial derivative $\frac{\partial}{\partial B}(S + D') = 1 - 2\alpha$, which is negative. Thus the maximum is achieved at $B = 0$. It is straightforward, if tedious, to then try all possible values of A and C to verify that $S + D' \leq 0$. (It's the case that $A = 0$ and $C = 1$ that requires $\alpha \geq 4/7$.)

Thus the Diminisher can ensure that the energy of the maximal independent set M —which is $|M|$ —is at most αn . QED

4 Minimum Value in Trees of Maximum Degree 3

It is unclear what the minimum value of the game is for trees of maximum degree at most 3. Perhaps it is even $3n/7 - O(1)$. Indeed, for trees of small orders, computer search shows that with Sweller going first the path is extremal. The path is not extremal when Diminisher goes first. For example, for order 11 there are four extremal graphs with $I_d = 4$: in each case, the graph is obtained from a disjoint P_7 and $K_{1,3}$ by adding an edge from a leaf of $K_{1,3}$ to a vertex of P_7 (an optimal move for Diminisher is to choose the center of the $K_{1,3}$).

A simple lower bound can be obtained by Sweller always choosing a leaf-vertex (or isolate). Thus every vertex she chooses dominates at most 2

vertices, while every vertex Diminisher chooses dominates at most 4 vertices, so that the value of the game is at least $2n/(4+2) - O(1)$. One can do slightly better:

Theorem 5 *For any tree T of order n with maximum degree at most 3, the value of the game with Sweller going first is at least $I_s(T) \geq 3n/8$.*

PROOF. We prove a slightly stronger statement for forests F of maximum degree at most 3. Let a be the number of isolates, b the number of components of K_2 , and $n' = n - a - 2b$. Then we claim the value of the game with Sweller going first is at least $\phi(F) = a + b + 3n'/8$.

The proof is by induction on n' . If $n' = 0$, then the value of the game is exactly $a + b$, as required. So assume $n' > 0$. We provide a strategy for Sweller. In each case, the worst that Diminisher can do for the analysis is to choose a vertex of degree 3 without creating a small component.

Consider a longest path P in the forest F . Say P is $v_1 v_2 \dots v_k$. If v_2 is incident with another end-vertex, then Sweller chooses vertex v_1 . Let $F' = F - \{v_1, v_2\}$. It follows that the value of the game is at least $2 + \phi(F') \geq 2 + (a + 1) + b + 3(n' - 7)/8 = \phi(F) + 3/8$.

So we may assume $k \geq 4$ and that v_2 has degree 2. If v_3 has a leaf neighbor, then Sweller chooses v_2 . By similar analysis, the value of the game is at least $2 + (a + 1) + b + 3(n' - 8)/8 = \phi(F)$.

If v_3 has degree 3, then its non- v_4 neighbor must (by above) be degree 2 with a leaf neighbor. Then Sweller chooses v_3 . By similar analysis, the value of the game is at least $2 + (a + 2) + b + 3(n' - 10)/8 = \phi(F) + 1/4$.

Finally, assume v_3 has degree 2. Then Sweller again chooses v_3 . By similar analysis, the value of the game is at least $2 + (a + 1) + b + 3(n' - 8)/8 = \phi(F)$. QED

5 Further Thoughts on Trees

For any fixed maximum degree Δ , one can readily show that the value of the game is at least $\Omega(n/\Delta)$. However, we are unable to find any tree T where $I_s(T) \leq 3n/7$.

One can also readily show that the value of the game is at most $n - \Omega(n/\Delta)$. However, we believe:

Conjecture 1 *For all trees T of order $n \geq 2$, the competition-independence number with Diminisher moving first is at most $I_d(T) \leq 3n/4$.*

If true, this conjecture is somewhat sharp, in that there are trees with $I_d(T) \geq 3n/4 - o(n)$. Construct a tree $T(a, b)$ with diameter 5 where the two central vertices have degree a and all their other neighbors have degree b , with a and b large. Then for Sweller's first move, if Diminisher hasn't chosen either of the centers then Sweller can choose one. So after the first two moves, Sweller can ensure that the graph has roughly half the vertices isolated and roughly half the vertices in $a - 1$ stars. Sweller can then get to choose a leaf in about half of the stars.

Finally, computer evidence suggests that in a tree both players always want to go first. This is in contrast to general graphs, where zugzwang is common. For example, in the 6-cycle both players want to go second. Indeed, by Theorem 1, if n is congruent to 1 or 6 modulo 7, then the value for Diminisher going first in the n -cycle is one more than the value for Sweller going first.

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